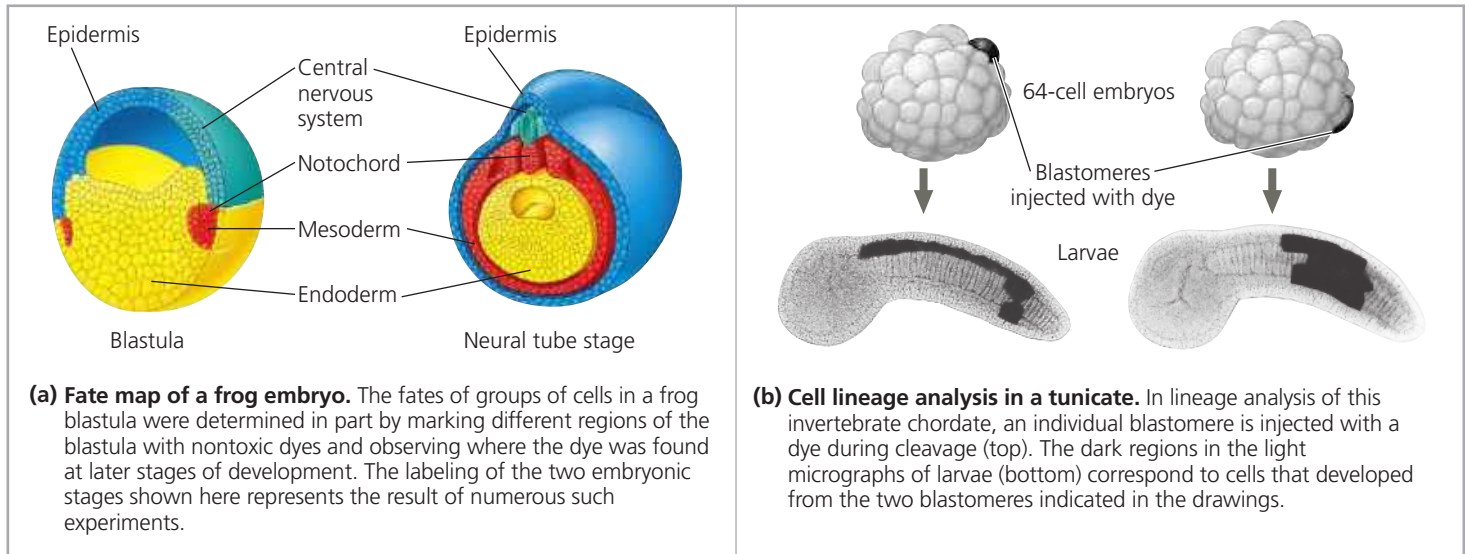


▼ **Figure 47.18 Fate mapping for two chordates.**



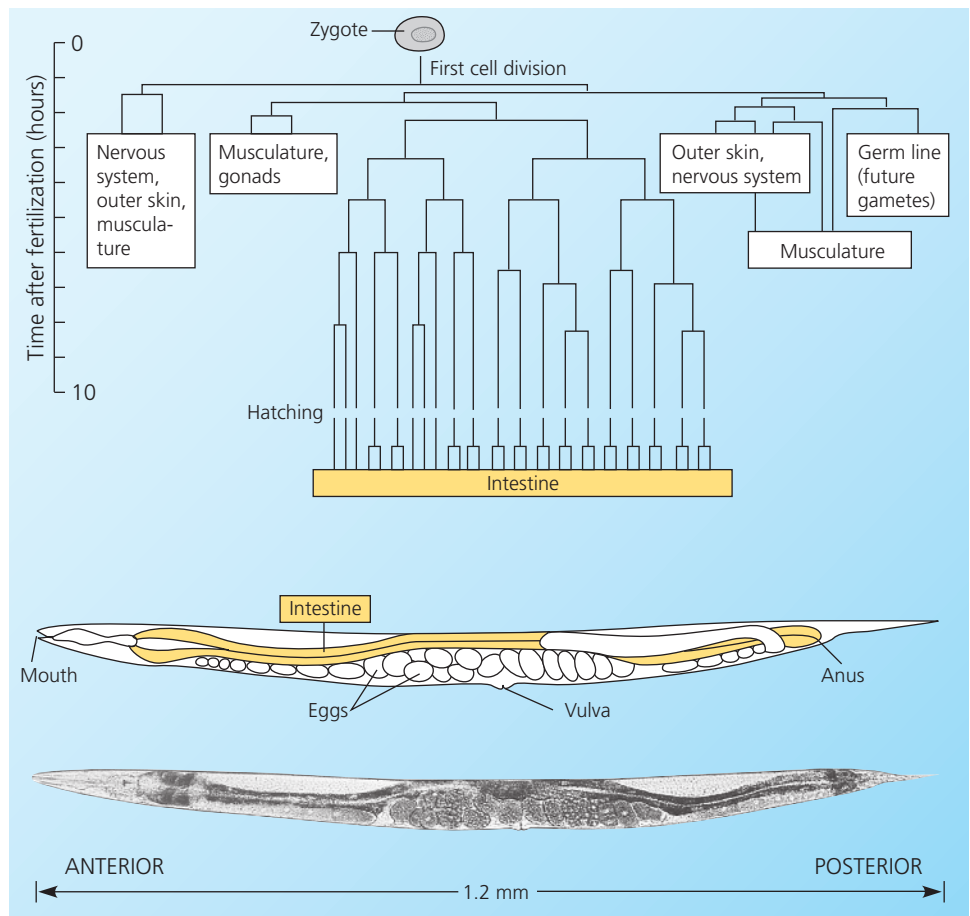
Fate Mapping

One way to trace the ancestry of embryonic cells is direct observation through the microscope. Such studies produced the first **fate maps**, diagrams showing the structures arising from each region of an embryo. In the 1920s, German embryologist Walther Vogt used this approach to determine where groups of cells from the blastula end up in the gastrula (**Figure 47.18a**). Later, researchers developed techniques that allowed them to mark an individual blastomere during cleavage and then follow the marker as it was distributed to all the mitotic descendants of that cell (**Figure 47.18b**).

A much more comprehensive approach to fate mapping has been carried out on the soil-dwelling nematode *Caenorhabditis elegans*, as shown in **Figure 47.19**. This roundworm is about 1 mm long, has a simple, transparent body with only a few types of cells, and develops into a mature adult hermaphrodite worm in only 3.5 days in the laboratory. These attributes allowed Sydney Brenner, Robert Horvitz, and John Sulston to determine the complete developmental history, or **lineage**, of every cell in *C. elegans*. They found that every adult hermaphrodite worm has exactly 959 somatic cells, which arise



▼ **Figure 47.19 Cell lineage in *Caenorhabditis elegans*.** The *C. elegans* embryo is transparent, making it possible for researchers to trace the lineage of every cell, from the zygote to the adult worm (LM). The diagram shows a detailed lineage only for the intestine, which is derived exclusively from one of the first four cells formed from the zygote.



VISUAL SKILLS The pattern of divisions is exactly the same in every *C. elegans* embryo. How many divisions of the fertilized egg give rise to the intestinal cell closest to the worm's mouth?